

SmartCoin: Enabling financial wellness for the underbanked

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SmartCoin began as a consumer lending platform on Google Cloud, enabling credit access for the underbanked in India. It has expanded as a wellness platform, offering a diversified portfolio of digital gold savings, insurance, credit, and more among others.



About SmartCoin

Incorporated in 2015, SmartCoin Financials Pvt. Ltd. is an app-based consumer lending platform with a vision to address the concerns of the vast underbanked in India.

Industries: Financial Services & Insurance

Location: India

(<https://cloud.google.com/compute/>)

Google Cloud results

- Scales analytics and machine learning models to serve millions of users across more than 19,000 postal codes across India without affecting performance
- Expands into savings and insurance offerings by saving on infrastructure overheads and focusing resources on products
- Safeguards and monitors access to sensitive data simply and effectively

Analysed
billions of
points
monthly



About Searce

Searce is a cloud-native tech consulting firm that empowers clients to modernize their businesses, leveraging Cloud, AI, & Analytics.

ABOUT SEARCE

Google Cloud (<https://cloud.google.com/>)

Searce is a cloud-native tech consulting

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Analytics

Cloud SQL (<https://cloud.google.com/sql/>)

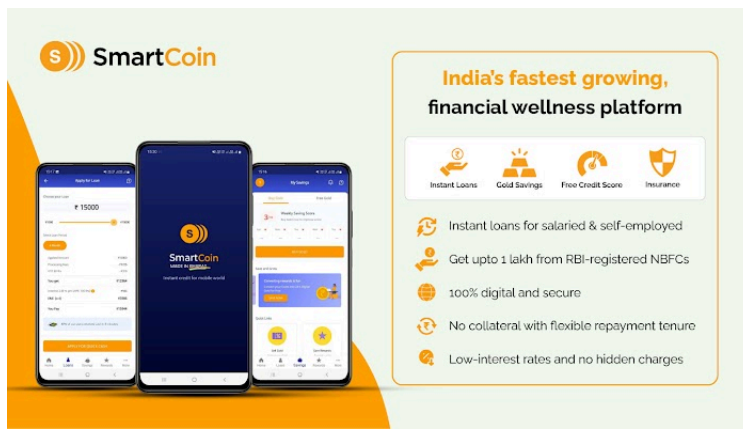
Compute Engine

(<https://cloud.google.com/compute/>)

with Identity and Access Management

In India, regular banking and credit facilities have traditionally been restricted to a well-connected, well-heeled, predominantly urban minority. While there have been huge improvements in recent years, according to the World Bank's 2021 Findex, 22% of Indians still have no access to a bank account. But the rise of Non-Banking Financial Companies (NBFCs), who specialize in lending small amounts of capital collateral-free, has helped the underbanked population gain access to vital avenues of finance, generating business and wealth creation in even the most rural areas.

While access to the traditional banking system may be restricted, smartphones have exploded in popularity in India, with more than 750 million users in 2022. That's where SmartCoin (<https://smartcoin.co.in/>), a consumer lending platform that partners with the Reserve Bank of India-registered non-banking financial companies (NBFCs), saw an opportunity. "We're trying to solve the problem of access to credit and financing using the power of data and technology," says Amit Chandel, CTO of SmartCoin. With its intuitive mobile app, SmartCoin aims to bring financial services to unbanked people.



SmartCoin app's features

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—Amit Chandel, CTO,
SmartCoin

Incorporated in 2015 by a team of IT veterans with experience in diverse industries, SmartCoin has grown to become one of India's foremost consumer lending platforms, with an infrastructure built for scale with Google Cloud (<https://cloud.google.com/>).

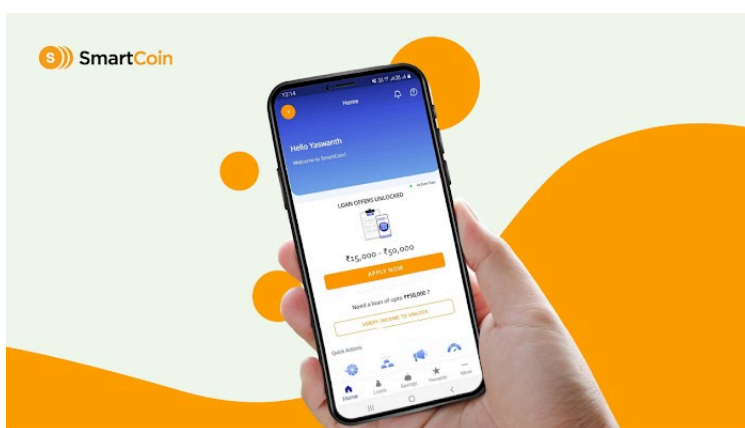
"I had previous experience with Google Cloud and knew it was just what we needed to start SmartCoin with," says Chandel. "It has the scalability, cost efficiency and reliability to help us grow rapidly and deliver the best possible service to our customers as we do so."

Finance without the friction

SmartCoin's app is designed to not only broaden access to financial services, but also make the experience as frictionless as possible. "That's a big differentiator between us and the banks," explains Chandel. "If you apply for a loan with a bank, you have to provide a huge volume of information, often physical documents, which many people simply don't have access to. We designed our application screens to be as easy to fill out as possible, while still maintaining our fiduciary responsibilities."

Users can register, fill out a simple questionnaire, provide KYC information digitally and complete their applications in minutes. While this is impressive enough, what makes SmartCoin a hit is how quickly this information is processed. By harnessing the power of machine learning and cutting-edge analytics, the platform can crunch through the data rapidly and tell its users if they are eligible for a loan as quickly as possible. "We have a hard deadline of 30 seconds to notify users of their eligibility," says Chandel.

Building an app like SmartCoin required striking a fine balance for the company. On the one hand, SmartCoin needed access to powerful resources in order to run its machine learning models and analytics at speed. On the other hand, SmartCoin is a startup with financial, workforce related and timebound limitations, so efficiency is also critical to its success.



SmartCoin app's easy UI

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—Amit Chandel, CTO,
SmartCoin

After deciding on Google Cloud for its infrastructure, SmartCoin chose to work with Google Cloud partner [Searce](https://searce.com/) (<https://searce.com/>). "Searce has been helpful for us in terms of getting additional support for any issues we've run into with Google Cloud and with competitive pricing," says Chandel. "We have also been able to get more strategic advice from their cloud experts."

SmartCoin uses Google Compute Engine (<https://cloud.google.com/compute>) to run its clusters, which can autoscale up or down as needed, requiring minimal input from technical staff. As well as providing services for the app, these clusters also host the company's machine learning solutions, which it has customized from open source tools. For the machine learning model training clusters, the company uses preemptible nodes to keep costs as low as possible. For its database management, the company uses Cloud SQL (<https://cloud.google.com/sql>) as well as Cloud Bigtable (<https://cloud.google.com/bigtable>) for NoSQL databases while BigQuery (<https://cloud.google.com/bigquery>) forms the data warehouse that makes SmartCoin as smart as it is.

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High quality financial services at scale

Over the past seven years, SmartCoin has gone from strength to strength. "With Google Cloud, scaling hasn't been a problem for us," says Chandel. "It's easy to go from thousands of users to millions. Today, we have nearly 13 million users across India and we crunch 140 billion data points every month."

One reason for SmartCoin's success has been its ability to stay lean and operationally efficient as it

has expanded. There is a strong focus on automation and to provide self-serve tools for internal teams. The major one being the internal ML-platform which has been developed in-house using open source technologies and Google Cloud infrastructure. The ML platform is leveraged across the board to enhance the product experience and to build industry leading fraud, risk, and collection models over non-traditional data sources. For cost efficiency, this platform makes full use of Google Cloud preemptible Virtual Machines during feature engineering and hyper-parameter optimization phase, and Bigtable API and BigQuery for feature store.

With the ease of use of the Google Cloud console and autoscaling abilities in Google Cloud, SmartCoin has been able to keep infrastructure overheads to a minimum while focusing resources on core business challenges and developing new features and products.



SmartCoin employees at work

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—Amit Chandel, CTO,
SmartCoin

SmartCoin started with a credit-first product strategy wherein customers were acquired through the microcredit hook. On the credit business, SmartCoin, enabled by Google Cloud has demonstrated an ability to acquire a unique and largely unaddressed customer base at low cost, underwrite them with time-tested models and engage them to drive high profitability. The credit hook is being leveraged by SmartCoin to strengthen

its expansion into further products including gold savings and insurance and effectively serve the entire financial lives of customers. These new products carry with them added compliance requirements but with [Google Cloud Security](https://cloud.google.com/security) (<https://cloud.google.com/security>) and [Identity and Access Management](https://cloud.google.com/iam) (<https://cloud.google.com/iam>), SmartCoin has been able to easily restrict and monitor access to sensitive information and keep its customers' data safe.

Moving forward, SmartCoin is planning for major growth in the near future and will lean just as heavily on Google Cloud as it has done since it launched. "We like using as much Google Cloud technology as we can," says Chandel. "What's great is that as we have grown over the years, we've seen Google Cloud products like BigQuery evolve as well."